

Functions: Composite & Inverse

ANSWER KEY



Domain and range

- Find the domain of $f(x) = \frac{1}{x-5}$.
The denominator cannot equal zero: $x \neq 5$. Domain: all real x except 5.
 - Find the domain of $g(x) = \sqrt{x-3}$.
The radicand must be non-negative: $x-3 \geq 0$, so $x \geq 3$.
 - Find the domain of $h(x) = \frac{1}{\sqrt{x-1}}$.
The radicand must be positive (it cannot be zero since it appears in the denominator): $x-1 > 0$, so $x > 1$.
 - Find the range of $f(x) = x^2 + 3$, for all real x .
Since $x^2 \geq 0$ for all real x , $f(x) = x^2 + 3 \geq 3$. Range: $f(x) \geq 3$.
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Evaluating functions

- If $f(x) = 2x^2 - 3x + 1$, find $f(-2)$.
 $f(-2) = 2(4) - 3(-2) + 1 = 8 + 6 + 1 = 15$.
 - If $f(x) = x^3 - 4x + 2$, find $f(3)$.
 $f(3) = 27 - 12 + 2 = 17$.
 - If $g(x) = \frac{x+3}{x-2}$, find $g(4)$.
 $g(4) = \frac{4+3}{4-2} = \frac{7}{2} = 3.5$.
 - If $h(x) = \sqrt{2x+7}$, find $h(9)$.
 $h(9) = \sqrt{2(9)+7} = \sqrt{25} = 5$.
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Composite functions

- Given $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $fg(x)$ (meaning $f(g(x))$).
 $fg(x) = f(x^2 - 3) = 2(x^2 - 3) + 1 = 2x^2 - 6 + 1 = 2x^2 - 5$.
- Using the same f and g , find $gf(x)$, and explain why $fg(x) \neq gf(x)$ in general.
 $gf(x) = g(2x + 1) = (2x + 1)^2 - 3 = 4x^2 + 4x + 1 - 3 = 4x^2 + 4x - 2$. Since $fg(x) = 2x^2 - 5$ is a different expression from $gf(x) = 4x^2 + 4x - 2$, composition of functions is generally not commutative — the order in which the functions are applied changes the result.
- Using the same $f(x) = 2x + 1$, find $ff(x)$ (meaning $f(f(x))$).
 $ff(x) = f(2x + 1) = 2(2x + 1) + 1 = 4x + 2 + 1 = 4x + 3$.

- 12 Given $f(x) = x^2 - 1$ and $g(x) = 2x + 3$, find $fg(x)$ (meaning $f(g(x))$).
- $$fg(x) = f(2x + 3) = (2x + 3)^2 - 1 = 4x^2 + 12x + 9 - 1 = 4x^2 + 12x + 8.$$

Finding an unknown in a composite function

- 13 Given $f(x) = 3x - 2$ and $g(x) = x + 4$, find the value of x such that $fg(x) = 13$.
- $$fg(x) = 3(x + 4) - 2 = 3x + 10. \text{ Set equal to 13: } 3x + 10 = 13, \text{ so } x = 1.$$
- 14 Using the same $f(x) = 3x - 2$ and $g(x) = x + 4$, find the value of x such that $gf(x) = 15$.
- $$gf(x) = g(3x - 2) = (3x - 2) + 4 = 3x + 2. \text{ Set equal to 15: } 3x + 2 = 15, \text{ so } 3x = 13, \text{ giving } x = \frac{13}{3}.$$
- 15 Given $f(x) = 2x + 5$ and $g(x) = 3x - 1$, find the value of x such that $fg(x) = 21$.
- $$fg(x) = 2(3x - 1) + 5 = 6x - 2 + 5 = 6x + 3. \text{ Set equal to 21: } 6x + 3 = 21, \text{ so } 6x = 18, \text{ giving } x = 3.$$
- 16 Given $f(x) = x + 6$ and $g(x) = 2x$:
- a Find $fg(x)$. $fg(x) = f(2x) = 2x + 6.$
 - b Solve $fg(x) = 20$ for x . $2x + 6 = 20$, so $2x = 14$, giving $x = 7.$

Inverse functions

- 17 Find $f^{-1}(x)$ for $f(x) = \frac{2x + 5}{3}$.
- $$\text{Let } y = \frac{2x + 5}{3}. \text{ Swap } x \text{ and } y: x = \frac{2y + 5}{3}. \text{ Solve: } 3x = 2y + 5, \text{ so } y = \frac{3x - 5}{2}. \text{ Thus } f^{-1}(x) = \frac{3x - 5}{2}.$$
- 18 Find $f^{-1}(x)$ for $f(x) = x^2 + 4$, for $x \geq 0$, and explain why the restriction $x \geq 0$ is needed.
- $$\text{Let } y = x^2 + 4: x^2 = y - 4, \text{ so } x = \sqrt{y - 4} \text{ (taking the positive root since } x \geq 0). \text{ Thus } f^{-1}(x) = \sqrt{x - 4}.$$
- The restriction is needed because $f(x) = x^2 + 4$ is not one-to-one over all reals (e.g. $f(2) = f(-2) = 8$), so it has no inverse unless the domain is restricted to make it one-to-one.
- 19 Find $f^{-1}(x)$ for $f(x) = \frac{4x - 1}{5}$.
- $$\text{Let } y = \frac{4x - 1}{5}. \text{ Swap } x \text{ and } y: x = \frac{4y - 1}{5}. \text{ Solve: } 5x = 4y - 1, \text{ so } y = \frac{5x + 1}{4}. \text{ Thus } f^{-1}(x) = \frac{5x + 1}{4}.$$
- 20 Find $f^{-1}(x)$ for $f(x) = \frac{x - 7}{2}$.
- $$\text{Let } y = \frac{x - 7}{2}. \text{ Swap } x \text{ and } y: x = \frac{y - 7}{2}. \text{ Solve: } 2x = y - 7, \text{ so } y = 2x + 7. \text{ Thus } f^{-1}(x) = 2x + 7.$$

The relationship between a function and its inverse graphically

- 21** If the point $(3, 11)$ lies on the graph of $y = f(x)$, state a point that must lie on the graph of $y = f^{-1}(x)$, and explain the general geometric relationship between the two graphs.
The point $(11, 3)$ must lie on $y = f^{-1}(x)$, since inverse functions reverse input-output pairs. Geometrically, the graph of f^{-1} is the reflection of the graph of f in the line $y = x$.
- 22** The graph of $y = f(x)$ passes through $(-2, 7)$. State the point that must lie on the graph of $y = f^{-1}(x)$.
Since inverse functions reverse input-output pairs, the point $(7, -2)$ must lie on $y = f^{-1}(x)$.
- 23** The graph of $y = f(x)$ has a y -intercept at $(0, 9)$. State the corresponding point that must lie on the graph of $y = f^{-1}(x)$.
Since $f(0) = 9$, the point $(0, 9)$ lies on $y = f(x)$. Reversing input and output, the point $(9, 0)$ must lie on $y = f^{-1}(x)$.

Range of a composite function

- 24** Given $f(x) = x^2$ with domain all reals, and $g(x) = x - 1$, find the range of $fg(x)$.
 $fg(x) = (x - 1)^2$. Since squaring any real number gives a value ≥ 0 , the range of $fg(x)$ is $fg(x) \geq 0$.
- 25** Given $f(x) = \sqrt{x}$ with domain $x \geq 0$, and $g(x) = x + 5$, find the range of $fg(x)$.
 $fg(x) = \sqrt{x + 5}$. As $x + 5$ takes all values ≥ 0 (for $x \geq -5$), the square root takes all values ≥ 0 , so the range of $fg(x)$ is $fg(x) \geq 0$.
- 26** Given $f(x) = \frac{1}{x}$ with domain $x \neq 0$, and $g(x) = x - 2$, find the range of $fg(x)$.
 $fg(x) = \frac{1}{x - 2}$. Since a fraction with a nonzero numerator can never equal zero, the range of $fg(x)$ is all real numbers except 0.
- 27** Given $f(x) = x^2$ with domain all reals, and $g(x) = 3 - x$:
- a** Find $fg(x)$. $fg(x) = f(3 - x) = (3 - x)^2$.
- b** State the range of $fg(x)$. Since squaring any real number gives a value ≥ 0 , the range of $fg(x)$ is $fg(x) \geq 0$.

A well-designed word problem: currency conversion functions

- 28** This question has two parts. A currency exchange converts US dollars to euros using $f(x) = 0.92x$, then charges a flat 3-euro service fee, giving the final amount $g(f(x))$ where $g(y) = y - 3$. (a) Find the amount of euros received for 200 dollars. (b) Find the inverse function of $h(x) = g(f(x))$, and use it to determine how many dollars must be exchanged to receive exactly 100 euros.
- (a) $f(200) = 0.92(200) = 184$. $g(184) = 184 - 3 = 181$ euros. (b) $h(x) = g(f(x)) = 0.92x - 3$. Solve for the inverse: $y = 0.92x - 3$, so $x = \frac{y + 3}{0.92}$, giving $h^{-1}(x) = \frac{x + 3}{0.92}$. For 100 euros:
 $h^{-1}(100) = \frac{103}{0.92} \approx 111.96$, so approximately 111.96 dollars must be exchanged.